

Local-Stochastic volatility model based on Schrödinger bridges

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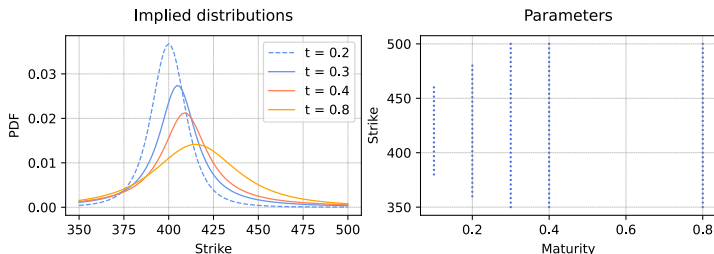
Motivation

The problem of volatility modeling

From the prices of the **options**, we can extract the implied distributions of the future returns of the asset $(S_t)_{t \geq 0}$

$$C(T, K) = E[(S_T - K)^+] \quad P(T, K) = E[(K - S_T)^+]$$

(Breeden-Litzenberger):
$$\frac{\partial^2 C}{\partial K^2}(K, \tau) = \varphi_{S_\tau}(x)$$



These distributions are not Gaussian, so it is necessary to use exotic processes to model the market.

Plan

1. Classical SV and LSV models.
2. Schrödinger Bridge construction.
3. Application to LSV models (Martingale Schrödinger bridge).
4. Practical implementation algorithm and extensions.

Volatility models: Stochastic volatility

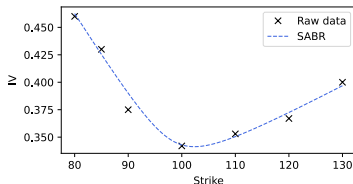
To capture the dynamics implied by the market prices, additional stochastic processes for variance (v_t) are used:

$$\begin{aligned}dS_t &= rS_t dt + v_t S_t dB_t^1 \\ dv_t &= a(v_t)dt + b(v_t)dB_t^2\end{aligned}$$

However, the parametric nature of SV models does not allow perfect fit.

SABR model (S.Hagan 2002):

$$\begin{aligned}dS_t &= v_t S_t^\beta dB_t^1 \\ dv_t &= \alpha v_t dB_t^2\end{aligned}$$



Analytic calibration to the market grid: MSE: $\min_{\theta} \frac{1}{N} \sum_{t,k} (C_{tk} - C_{tk}^{\theta})^2$

Volatility models: Local volatility

Another approach to volatility modeling is to introduce a nonparametric *Local volatility function* $\sigma(t, S_t)$, that reproduces the market option prices:

$$dS_t = rS_t dt + \sigma_{Dup}(t, S_t) S_t dB_t$$

There exists an explicit formula for such a function introduced in (Dupire, 1994):

$$\sigma_{Dup}(K, T) = \sqrt{\frac{\frac{\partial C}{\partial T} + rK \cdot \frac{\partial C}{\partial K}}{\frac{1}{2}K^2 \cdot \frac{\partial^2 C}{\partial K^2}}}$$

1. Calibration requires $C(K, T)$ - continuous and differentiable in both arguments, while only a discrete set of prices are available $C(K_i, T_i)$
2. Provides the incorrect dynamics for implied volatility.

Volatility models: LSV

To obtain the perfect fit and suitable dynamics these methods are combined. Non-parametric function $\sigma(t, S_t)$ is added on top of the stochastic volatility process (v_t) :

$$\begin{aligned}dS_t &= rS_t dt + \sigma(t, S_t)\psi(v_t)S_t dB_t^1 \\ dv_t &= a(v_t)dt + b(v_t)dB_t^2\end{aligned}$$

The function $\sigma(t, S_t)$ includes a Dupire-like equation, though LSV models still require fill interpolation of option prices:

$$\sigma(t, S_t) = \sqrt{\frac{\sigma_{Dup}(t, S_t)}{\mathbb{E}(\psi^2(v_t)|S = S_t)}}$$

Idea: *Modify the dynamics of the process to match the distributions only at particular time points.*

Schrödinger problems

In 1931 Schrödinger addressed a problem that is translated in modern terms as follows:

1. Dynamic problem (Bridge): Let R be a measure on the path-space $C[0, 1]$. Consider the family of measures, with projection μ at $t = T$

$$\mathcal{M} = \{\mathbb{P} \ll R : \pi_T \circ \mathbb{P} = \mu\}$$

$$\min_{\mathbb{P} \in \mathcal{M}(\mu)} H(\mathbb{P}|R) = \min_{\mathbb{P} \in \mathcal{M}(\mu)} \mathbb{E}^{\mathbb{P}} \left[\ln \left(\frac{d\mathbb{P}}{dR} \right) \right]$$

The idea of the dynamic problem is to find the most similar process to the reference, but with a fixed distribution at $t = 1$.

2. Static problem: Let R be a measure on the path-space $C[0, 1]$. Consider a projection of reference measure:

$$\min_{\pi} H(\pi|R_{01}), \quad \pi \in \Pi(X_0, X_1), \quad R_{01} = (X_0, X_1)_{\#}R$$

To apply the Schrödinger problems to stochastic volatility models we first solve a simple problem:

Problem 1 (Reference process: Brownian motion): Dynamic problem with Winner reference measure:

$$\min_{\mathbb{P} \in \mathcal{M}(\mu)} H(\mathbb{P} | \mathbb{P}^W) \quad \mathcal{M} = \{\mathbb{P} \sim \mathbb{P}^W : X_0 \sim^{\mathbb{P}} \delta_0, X_T \sim^{\mathbb{P}} \mu\}$$

Then we change a reference measure to a stochastic volatility model:

Problem 2 (Reference process: SV model): Dynamic problem with $\mathbb{P}^0$ is generated by SV model and martingale constraint:

$$\min_{\mathbb{P} \in \mathcal{M}(\mu)} H(\mathbb{P} | \mathbb{P}^0) \quad \mathcal{M} = \{\mathbb{P} \sim \mathbb{P}^0 : S_0 \sim^{\mathbb{P}} \delta_0, \\ S_T \sim^{\mathbb{P}} \mu, \quad \mathbb{E}(S_t | \mathcal{F}_u) = S_u\}$$

Schrödinger bridge

Original approach. One Marginal

Problem 1: First consider a problem with Winner reference measure:

$$\text{SP}(\delta_0, \mu) := \min_{\mathbb{P} \in \mathcal{M}(\mu)} H(\mathbb{P} | \mathbb{P}^W) \quad \mathcal{M} = \{\mathbb{P} \sim \mathbb{P}^W : X_0 \sim^{\mathbb{P}} \delta_x, X_T \sim^{\mathbb{P}} \mu\}$$

Solution 1: Consider a strictly concave optimization:

$$P_1 = \sup_{f \in L^1(\mu)} \{-\mathbb{E}^\mu(f) - \ln \mathbb{E}[e^{-f(W_T)} | W_0 = x]\}$$

Then f^* is a unique solution. And the diffusion:

$$dX_t = \partial_x \ln \mathbb{E}[e^{-f^*(W_T)} | W_t = X_t] dt + dW_t$$

has a given marginal $X_T \sim^{\mathbb{P}} \mu$

The optimal process has a form of diffusion with path-dependent drift!

Application

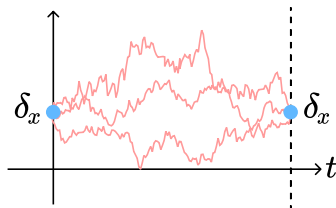
Corollary: Applying the variational optimization we obtain the optimality condition for the potential f^* and plugging it into the process get the representation:

$$dX_t = \frac{\int \frac{(y - X_t)}{T - t} f(y, X_t) \mu(dy)}{\int f(y, X_t) \mu(dy)} dt + dW_t \quad X_0 = \delta_x, \quad X_T \sim \mu$$

We skip the proof for now

Example:

$$dX_t = \frac{(x - X_t)}{T - t} dt + dW_t \quad X_0 = \delta_x, \quad X_T \sim \delta_x$$



The framework gives a common result in a simple case.

Schrödinger bridge

Girsanov Theorem

Proof: We are looking for the measure $\mathbb{P}$ that is equivalent to $\mathbb{P}^W$. According to the Girsanov theorem [ref](#) the process X that corresponds to the measure $\mathbb{P}$ can be represented with the SDE under $\mathbb{P}$:

$$dX_t = \lambda_t dt + dW_t$$

$$\frac{d\mathbb{P}^W}{d\mathbb{P}} \Big|_{\mathcal{F}_t} = \exp \left(- \int_0^t \lambda_s dW_s - \frac{1}{2} \int_0^t \lambda_s^2 ds \right)$$

The form of the Radon-Nykodym derivative perfectly fits the form of cost function $H(\mathbb{P}|\mathbb{P}^W)$ and gives a simple reformulation:

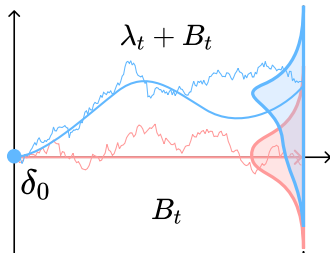
$$H(\mathbb{P}|\mathbb{P}^W) = \mathbb{E}^{\mathbb{P}} \left[\ln \left(\frac{d\mathbb{P}}{d\mathbb{P}^W} \right) \right] = \frac{1}{2} \mathbb{E}^{\mathbb{P}} \int_0^T \lambda_s^2 ds$$

Schrödinger bridge

Reformulation

The Schrödinger problem is reduced to the stochastic control problem:

$$\text{SP}(\delta_0, \mu) := \min_{\mathbb{P} \in \mathcal{M}} H(\mathbb{P} | \mathbb{P}^W) = \min_{\lambda_t: X_0 \sim \delta_0, X_T \sim \mu} \frac{1}{2} \mathbb{E}^{\mathbb{P}} \int_0^T \lambda_s^2 ds$$



Still, the domain of optimization is complicated, so we need to implement the constraints in a simpler form.

Schrödinger bridge

Solution

(1) To solve the optimization the Lagrange multiplier is added to capture the restriction on the final marginal:

$$\inf_{\mathbb{P} \in \mathcal{M}(\mu)} \frac{1}{2} \mathbb{E}^{\mathbb{P}} \left(\int_0^T \lambda_s^2 ds \right) = \sup_{f \in L^1(\mu)} \inf_{(\lambda_s)} \frac{1}{2} \mathbb{E}^{\mathbb{P}} \left(\int_0^T \lambda_s^2 ds \right) + \mathbb{E}^{\mathbb{P}} f(X_T) - \mathbb{E}^{\mu} f$$

$$\mathbb{E}^{\mathbb{P}} f(X_T) - \mathbb{E}^{\mu} f = 0 \text{ iff } X_T \sim \mu$$

(2) The optimization becomes: $\sup_{f \in L^1(\mu)} (u(0, X_0) - \mathbb{E}^{\mu}[f])$

$$u(t, x) = \inf_{\lambda} \mathbb{E}^{\mathbb{P}} \left(\int_0^T \frac{1}{2} \lambda_s^2 ds + f(X_T) \middle| X_t = x \right)$$

And $u(t, x)$ corresponds to a solution for HJB equation **HJB**:

$$\partial_t u + \frac{1}{2} \partial_x^2 u + \inf_{\lambda} \left(\frac{1}{2} \lambda^2 + \lambda \partial_x u \right) = 0, \quad u(T, x) = f(x)$$

Schrödinger bridge

By optimizing the equation over λ we first get an optimal control and second the solution $u(t, x)$ under this control:

$$\partial_t u + \frac{1}{2} \partial_x^2 u + \inf_{\lambda} \left(\frac{1}{2} \lambda^2 + \lambda \partial_x u \right) = 0, \quad u(T, x) = f(x)$$

$$\lambda^* = \partial_x u, \quad u(t, x) = -\ln \mathbb{E}[e^{-f(W_T)} | W_t = x]$$

Plugging it into the initial problem we get:

$$P_1 = \sup_{f \in L^1(\mu)} (-\mathbb{E}^\mu[f] - \ln \mathbb{E}[e^{-f(W_T)} | W_0 = x])$$

Let the result of optimization be f^* , the final answer to the problem is:

$$dX_t = \partial_x (\ln \mathbb{E}[e^{-f^*(W_T)} | W_t = X_t]) dt + dW_t \quad X_0 = \delta_x, \quad X_T \sim \mu$$

□

(X_t) can be simulated straightforward using Euler Scheme, or using the expression for Girsanov exponent.

Schrödinger bridge

Two Marginals

The proof extends to the case with multi marginals and for the more complicated processes!

$$\min_{\mathbb{P} \in \mathcal{M}(\mu)} H(\mathbb{P} | \mathbb{P}^W) \quad \mathcal{M} = \{\mathbb{P} \sim \mathbb{P}^W : X_1 \sim^{\mathbb{P}} \mu_1, X_2 \sim^{\mathbb{P}} \mu_2\}$$

Consider a strictly concave optimization:

$$P_1 = \sup_{f_1, f_2} \{-E^{\mu_1}(f_1) - E^{\mu_2}(f_2) - \ln E^{\mathbb{P}}[e^{-f_1(W_1) - f_2(W_2)} | W_0 = x]\}$$

Then f^* is a unique solution. And the diffusion:

$$dX_t = \partial_x \ln E^{\mathbb{P}}[e^{-f_1^*(W_1) - f_2^*(W_2)} | W_t = X_t] dt + dW_t$$

has a given marginal $X_1 \sim^{\mathbb{P}} \mu_1, X_2 \sim^{\mathbb{P}} \mu_2$

Martingale Schrödinger bridge

Application to LSV

The Schrödinger bridge framework can be used when the reference measure is given with a more complicated process.

Consider a calibrated stochastic volatility model. *(It does not fit perfectly to the market prices)*

$$\begin{aligned}dS_t &= S_t v_t dW_t \\ dv_t &= b(v_t)dt + \sigma(v_t)(\rho dW_t + \sqrt{1 - \rho^2} dW_t^\perp)\end{aligned}$$

Denote the measure $\mathbb{P}^0$ on $C[0, T] \times C[0, T]$ generated by this process as a reference measure for the two-dimensional Dynamic Schrodinger problem.

We find a process which is the most similar to $\mathbb{P}^0$ with respect to $H(\mathbb{P}|\mathbb{P}^0)$ but has a required marginals.

Martingale Schrödinger bridge

Application to LSV

Problem 2: Consider a reference measure on $(C[0, T])^2$:

$$\begin{aligned}\mathbb{P}^0 : \quad dS_t &= S_t v_t dW_t \\ dv_t &= b(v_t)dt + \sigma(v_t)(\rho dW_t + \sqrt{1 - \rho^2} dW_t^\perp)\end{aligned}$$

Solve a Dynamic Schrödinger problem:

$$\min_{\mathbb{P} \in \mathcal{M}(\mu)} H(\mathbb{P} | \mathbb{P}^0)$$

$$\mathcal{M} = \{\mathbb{P} \sim \mathbb{P}^0 : S_0 \sim^{\mathbb{P}} \delta_0, \quad S_T \sim^{\mathbb{P}} \mu, \quad \mathbb{E}(S_t | \mathcal{F}_u) = S_u\}$$

for simplicity, we analyze the problem with a trivial starting measure, however the solution extends to the case with multi-marginals.

Note that we additionally require the S_t to be a martingale .

Martingale Schrödinger bridge

Solution Consider a strictly concave optimization:

$$P_1 = \sup_{f, \Delta} \left\{ -\mathbb{E}^\mu[f] - \ln \mathbb{E}^{\mathbb{P}^0} [e^{-f(S_T) - \int_0^T \Delta_s dS_s}] \right\} =$$

Then the optimal f^* and Δ^* characterize the solution for the problem:

$$dS_t = S_t v_t dW_t \quad dZ dW = \rho dt$$

$$dv_t = b(v_t) + (1 - \rho^2) \sigma(v_t)^2 \partial_v \ln \mathbb{E}^{\mathbb{P}^0} [e^{-f_1^*(S_{t_1}) - \int_0^{t_1} \Delta_s^* dS_s} | S_t, v_t] dt + \sigma(v_t) dZ_t$$

1) $S_T \sim \mu$

2) $\Delta_t^* = -\left(\partial_s + \rho \frac{\sigma(v_t)}{S_t v_t} \partial_v\right) u(t, S_t, v_t)$ where u is a solution for the PDE:

$$\partial_t u + \mathcal{L}^0 u - \frac{1}{2} (1 - \rho^2) (\sigma(v) \partial_v u)^2 = 0 \quad u(T, s, v) = f_1^*(s)$$

Similarly to the last case the optimal f , Δ controll the final marginal μ and martingale constraint.

Martingale Schrodinger bridge

Proof Again, the Girsanov theorem is applicable in two dimensions, so the equivalent measure $\mathbb{P}$ is characterized by the two drift coefficients $\lambda^{(1)}, \lambda^{(2)}$, s.t. under $\mathbb{P}^0$:

$$dS_t = S_t v_t (\lambda_t^{(1)} dt + dW_t)$$

$$dv_t = b(v_t) dt + \sigma(v_t) (\rho(\lambda_t^{(1)} dt + dW_t) + \sqrt{1 - \rho^2} (\lambda_t^{(2)} dt + dW_t^\perp))$$

The Radon-Nycodm derivative, again gives a simple expression for relative entropy:

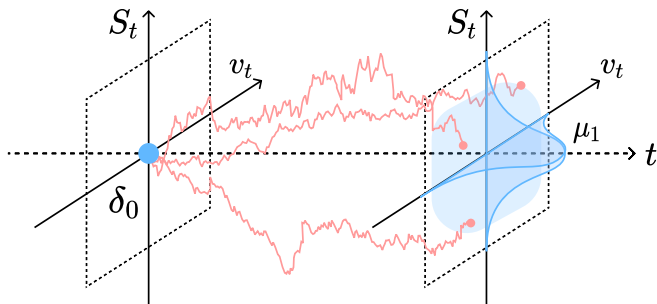
$$\frac{d\mathbb{P}^0}{d\mathbb{P}} \Big|_{\mathcal{F}_t} = \exp \left(-\int_0^t \lambda_s^{(1)} dW_s - \frac{1}{2} \int_0^t (\lambda_s^{(1)})^2 ds \right) \exp \left(-\int_0^t \lambda_s^{(2)} dW_s^\perp - \frac{1}{2} \int_0^t (\lambda_s^{(2)})^2 ds \right)$$

$$H(\mathbb{P}|\mathbb{P}^0) = \mathbb{E} \left[\frac{1}{2} \int_0^T (\lambda_s^{(1)})^2 ds + \frac{1}{2} \int_0^T (\lambda_s^{(2)})^2 ds \right]$$

Again we end up with the stochastic controll problem.

Illustration

$$H(\mathbb{P}|\mathbb{P}^0) = \min_{\lambda_t^{(1)}, \lambda_t^{(2)}: S_0 \sim \delta_0, S_T \sim \mu} \mathbb{E} \left[\frac{1}{2} \int_0^T (\lambda_s^{(1)})^2 ds + \frac{1}{2} \int_0^T (\lambda_s^{(2)})^2 ds \right]$$



Martingale Schrodinger bridge

Stochastic control framework

The optimization can be done over the controls $\lambda_s^{(1)}, \lambda_s^{(2)}$ with the martingale constraint, which can be implemented via integral Lagrange multiplier, together with the indicator of the final marginal:

$$\begin{aligned} \inf_{\mathbb{P} \in \mathcal{M}} H(\mathbb{P} | \mathbb{P}^0) &= \\ &= \sup_{f, \Delta} \inf_{\lambda^{(1)}, \lambda^{(2)}} \sum_1^2 \mathbb{E}^{\mathbb{P}} \left[\frac{1}{2} \int_0^T (\lambda_s^{(i)})^2 ds \right] + \mathbb{E}^{\mathbb{P}} f_1(S_T) - \mathbb{E}^{\mu}[f] + \mathbb{E} \int_0^T \Delta_s dS_s \end{aligned}$$

The process S_t is a martingale if and only if : $\mathbb{E} \left[\int_0^1 \Delta_s dS_s \right] = 0$

This formulation can be rewritten with:

$$P_1 = \sup_{f \in L^1(\mu)} (-\mathbb{E}^{\mu}[f] + u(0, S_0, v_0))$$

Where $u(t, s, v)$ is a solution for 2d HJB PDE $\rightarrow$

Martingale Schrodinger bridge

Two dimensional HJB

$$u(t, s, v) = \inf_{\lambda^{(1)}, \lambda^{(2)}} \sum_1^2 \mathbb{E}^{\mathbb{P}} \left[\frac{1}{2} \int_0^T (\lambda_s^{(i)})^2 ds + f(S_T) + \int_0^T \Delta_s dS_s | S_t, a_t \right]$$

$$\begin{aligned} & \partial_t u + \mathcal{L}^0 u \\ & \inf_{\lambda^{(1)}, \lambda^{(2)}} \sup_{\Delta \in \mathbb{R}} \lambda^{(1)} s v \partial_s u + \lambda^{(1)} \rho \sigma(v) \partial_v u + \lambda^{(2)} \sqrt{1 - \rho^2} \sigma(v) \partial_v u \\ & \frac{(\lambda^{(1)})^2}{2} + \frac{(\lambda^{(2)})^2}{2} + \lambda^{(1)} s v \Delta \end{aligned}$$

$\mathcal{L}^0 u$ is an Ito generator for the process with $\lambda^{(i)} = 0$

Solving the equation for the contrals $\lambda^{(1)}, \lambda^{(2)}, \Delta$:

$$\lambda^{(1)*} = 0$$

$$\lambda^{(2)*} = -\sqrt{1 - \rho^2} \sigma(v_t) \partial_v u$$

$$\Delta^* = -(\partial_s + \rho \frac{\sigma(v_t)}{S_t v_t} \partial_v) u$$

Martingale Schrodinger bridge

Solution

Under the optimal controls, PDE has a form:

$$\partial_t u + \mathcal{L}^0 u - \frac{1}{2}(1 - \rho^2)(\sigma(a)\partial_v u)^2 = 0, \quad u(T, s, v) = f(s)$$

The solution can be written back as a stochastic control problem:

$$u(t, s, v) = \sup_{\Delta} -\ln \mathbb{E}^{\mathbb{P}^0} [e^{-f(S_T) - \int_t^T \Delta_s dS_s} | S_t, a_t]$$

□

Using this result we can derive an algorithm for pricing with the Schrödinger LSV

→

Practical implementation

Finally, if we get the solution f^* for the optimization problem:

$$P_1 = \sup_f \left\{ -E^\mu[f] - u(0, S_0, v_0) \right\} =$$

Where u is a solution of PDE:

$$\partial_t u + \mathcal{L}^0 u - \frac{1}{2}(1 - \rho^2)(\sigma(v)\partial_v u)^2 = 0 \quad u(T, s, v) = f(s)$$

We will obtain the expression for "martingale control process" Δ^* :

$$\Delta_t^* = -\left(\partial_s + \rho \frac{\sigma(v_t)}{S_t v_t} \partial_v\right) u(t, S_t, v_t)$$

f^* and Δ^* are all the ingredients for the calculation of the Girsanov exponent:

$$\frac{d\mathbb{P}^0}{d\mathbb{P}} \Big|_{\mathcal{F}_t} = \exp \left(- \int_0^t \lambda_s^{(2)} dW_s^\perp - \frac{1}{2} \int_0^t (\lambda_s^{(2)})^2 ds \right)$$

Practical implementation

The final algorithm for implementation of the model:

1. Simulate the path of the SV model under measure $\mathbb{P}^0$:

$$dS_t = S_t v_t dW_t$$

$$dv_t = b(v_t)dt + \sigma(v_t)(\rho dW_t + \sqrt{1 - \rho^2} dW_t^\perp)$$

2. Solve the optimization, using the Newton gradient method, and find optimal f^* :

$$P_1 = \sup_{f \in L^1(\mu)} \{-E^\mu(f) - u(t, s, v)\}$$

$$\partial_t u + \mathcal{L}^0 u - \frac{1}{2}(1 - \rho^2)(\sigma(a)\partial_a u)^2 = 0, \quad u(T, s, a) = f(s)$$

Each step of the algorithm will calculate full PDE with final condition $f_1(s)$

3. Using the optimal $u(t, s, v)$ find a process: $\Delta^* = -(\partial_s + \rho \frac{\sigma(v_t)}{S_t v_t} \partial_v)u$
4. Plug the solution into the Girsanov exponent and calculate the value of the instruments in this LSV model:

$$E^{\mathbb{P}}[F] = E^{\mathbb{P}^0} \left[F \frac{d\mathbb{P}}{d\mathbb{P}^0} \right]$$

Example: Options on Variance

The framework can also be applied for calibration to exotic instruments:

$$dv_t = b(v_t)dt + \sigma(v_t)dZ_t^0$$

Let the realized quadratic variation be: $V_t = \langle \ln S \rangle_t = \int_0^t v_s^2 ds$. The option contract written on this value has a payoff:

$$C_\mu^{mkt}(k) = \mathbb{E}^\mathbb{P}[(V_T - K)^+] \Rightarrow V_t \sim \mu = \partial_K^2 C(K)$$

Proposition: Consider a strictly concave optimization:

$$P_1 = \sup_{f \in L^1(\mu)} \{-\mathbb{E}^\mu(f) - \ln \mathbb{E}^\mathbb{P}[e^{-f(V_T)}]\}$$

Then f^* is a unique solution. And the diffusion:

$$dv_t = (b(v_t) + \sigma^2(v_t)\partial_a \ln \mathbb{E}^{\mathbb{P}^0}[e^{-f^*(V_T)}|v_t, V_t])dt + \sigma(v_t)dW_t$$

has a given marginal $\int_0^t v_s^2 ds. \sim^\mathbb{P} \mu$

Possible extentions

Martingale Schrodinger Bridge:

$$\begin{aligned} \text{SP}(\delta_0, \mu) := \min_{M_0 \sim \delta_0, M_1 \sim \mu} & \frac{1}{2} \mathbb{E} \left[\int_0^1 \lambda_t^2 dt \right] & dM_t = \xi_s dB_s \\ & d\xi_t = \lambda_t dt + dB_s \end{aligned}$$

Martingale Benamou-Brenier:

$$\text{MT}(\delta_0, \mu) : \min_{M_0 \sim \delta_0, M_1 \sim \mu} \mathbb{E} \left[\int_0^1 (\sigma_t - 1)^2 dt \right] \quad dM_t = \sigma_s dB_s$$

Mixed problem:

$$\begin{aligned} \text{Mix}(\delta_0, \mu) : \min_{M_0 \sim \delta_0, M_1 \sim \mu} & \mathbb{E} \left[\int_0^1 L(\lambda_t, \sigma_t) dt \right] & dM_t = \sigma_s \xi_s dB_s \\ & d\xi_t = \lambda_t dt + dB_s \end{aligned}$$

Literature

-  H. Pham "Continuous-time Stochastic Control and Optimization with Financial Applications"
-  J. Gatheral "The Volatility surface"
-  P. Henry-Labordere "From (martingale) Schrödinger Bridges to a new class of Stochastic Volatility Models." (2019)s

Appendix A

Implied volatility

Definition. Implied volatility σ_{im} is a parameter that equates the market price of an option C^m to the theoretical price given by the Black-Scholes formula.

$$C^m(K, \tau) = C_{BS}(K, \tau, \sigma_{im}(K, \tau))$$

Definition. The volatility surface is a function that maps the option parameters to implied volatility

$$\text{IVS} : (K, \tau) \rightarrow \sigma_{im}(K, \tau)$$

where $\tau = T - t$ is the time to maturity and K is strike.

Appendix B

Girsanov theorem

Theorem (Girsanov). $(W_t)_{t \geq 0}$. Let $(\lambda_t)_{t \geq 0}$ be an adapted process such that: $\mathbb{E} \exp \left(\frac{1}{2} \int_0^T \lambda_s^2 ds \right) < \infty$. Define the process $(Z_t)_{t \geq 0}$

$$Z_t = \exp \left(- \int_0^t \theta_s dW_s - \frac{1}{2} \int_0^t \theta_s^2 ds \right).$$

Define a new probability measure $\mathbb{Q}$ by $\frac{d\mathbb{Q}}{d\mathbb{P}} = Z_T$. Under $\mathbb{Q}$, the process $W_t^{\mathbb{Q}}$ is a Brownian motion:

$$W_t^{\mathbb{Q}} = W_t + \int_0^t \theta_s ds$$

Appendix C

Hamilton-Jacobi-Bellman equations

Recall the framework of Hamilton Jacobi Bellman equations:

$$dX_t = b(\alpha_t, t, X_t)dt + \sigma(\alpha_t, t, X_t)dW_t$$

$$u(t, x) = \inf_{\alpha_t} \mathbb{E} \left[\int_0^1 L(\alpha, t, X_t)dt + f(X_t) \middle| X_t = x \right]$$

$$\partial_t u + \inf \{ b(\alpha, t, x) \partial_x u + \frac{1}{2} \sigma^2 \partial_{xx} u \} = 0$$